

# Donghae Seo

Postdoctoral Researcher / KAIST

[donghaeseo.github.io](https://donghaeseo.github.io) | [Google Scholar](#) | [ORCID: 0009-0005-3455-2430](#)

## Current Position

**Postdoctoral Researcher** Sep 2026 – Present  
Natural Science Research Institute, Korea Advanced Institute of Science and Technology (KAIST)  
Daejeon, South Korea | Alternative Military Service (Expected completion: Aug 2027)

## Education

**Ph.D. in Physics** Mar 2021 – Aug 2026  
Pohang University of Science and Technology (POSTECH), Pohang, South Korea  
Advisors: Prof. Gil Young Cho (Co-advised by Prof. Hee-Cheol Kim)  
Thesis: *Classification of Topological Orders and Its Applications*

**B.S. in Physics** Mar 2017 – Feb 2021  
Pohang University of Science and Technology (POSTECH), Pohang, South Korea

## Research Experience

**Visiting Graduate Researcher** Jun 2024 – Jul 2024  
Cavendish Laboratory, University of Cambridge, Cambridge, UK  
Host: Prof. Robert-Jan Slager  
Conducted collaborative research on noninvertible duality [*Physical Review B* **111**, 115112 (2025)].

## Publications & Preprints

\* Equal contributions    † Corresponding author

### Preprints

- [8] *Wavefunctions for Anyon Superconductors*  
D. Seo\*, T. Lee\*, and G. Y. Cho† | [arXiv:2609.04187](#)
- [7] *A Unified Categorical Description of Quantum Hall Hierarchy and Anyon Superconductivity*  
D. Seo, T. Lee, G. Y. Cho† | [arXiv:2602.03848](#)

### Published and Accepted Papers

- [6] *U(2) Chern-Simons-Ginzburg-Landau theory of fractional quantum Hall hierarchies*  
T. Lee, G. Y. Cho†, D. Seo† | *Physical Review B*, accepted as a Letter (2026) | Corresponding author
- [5] *Superconductivity proximate to non-Abelian fractional spin Hall insulator in twisted bilayer MoTe<sub>2</sub>*  
C.-E. Ahn\*, D. Seo\*, G. Lee, Y. Kim, G. Y. Cho† | *Nature Communications* **17**, 7005 (2026) | Co-first author
- [4] *Most Two-Dimensional Bosonic Topological Orders Forbid Sign-Problem-Free Quantum Monte Carlo Simulation: Nonpositive Gauss Sum as an Indicator*  
D. Seo, M. You, H.-C. Kim†, and G. Y. Cho† | *Physical Review Letters* **135**, 146602 (2025)
- [3] *Noninvertible duality and symmetry topological order of one-dimensional lattice models with spatially modulated symmetry*  
D. Seo, G. Y. Cho†, and R.-J. Slager | *Physical Review B* **111**, 115112 (2025)
- [2] *Modular extension of topological orders from congruence representations*  
D. Seo\*, M. You\*, G. Y. Cho†, and H.-C. Kim† | *Physical Review B* **109**, 155105 (2024)
- [1] *Classification of fermionic topological orders from congruence representations*  
G. Y. Cho†, H.-C. Kim†, D. Seo†, M. You† | *Physical Review B* **108**, 115103 (2023) | Authors listed in alphabetical order

## Talks & Presentations

---

### Seminars

- [7] *A Unified Approach to Quantum Hall Hierarchy and Anyon Superconductivity*  
Condensed Matter Theory Group, Massachusetts Institute of Technology Jul 2026
- [6] *A Unified Approach to Quantum Hall Hierarchy and Anyon Superconductivity*  
Department of Physics, Yale University Jul 2026
- [5] *Classification of Topological Orders and Its Applications*  
Cavendish Laboratory, University of Cambridge Jun 2024

### Conference Presentations

- [4] *Classification of Fermionic Topological Orders using Congruence Representations*  
2023 KPS Spring Meeting, Daejeon, South Korea Apr 2023  
Outstanding Presentation Award
- [3] *Classification of Fermionic Topological Orders from Congruence Representations*  
APS March Meeting 2023, Las Vegas, NV, USA Mar 2023
- [2] *Algebraic Theory of Anyons*  
Frontiers of Theoretical Physics 2022, Seogwipo, South Korea Nov 2022
- [1] *Classification of Non-Unitary MTC and Topological Phases*  
APS March Meeting 2022, Chicago, IL, USA Mar 2022

## Awards & Fellowships

---

- Byung Il Min Solid State Theory Paper Award, POSTECH** 2026  
One recipient per year, selected from among all graduate students and postdoctoral researchers
- Outstanding Paper Award, POSTECH** 2026  
Awarded for the year's top graduate student publications
- POSTECHIAN Fellowship, POSTECH** 2025  
Highly competitive university-wide fellowship awarded to a select number of students
- Outstanding Presentation Award, The Korean Physical Society** 2023  
Awarded for an outstanding oral presentation at the 2023 KPS Spring Meeting
- BK21 Interdisciplinary Scholar Fellowship, POSTECH** 2021–2024  
One recipient in the annual physics cohort; provides a full stipend

## Teaching & Mentoring

---

- Undergraduate Research Supervision** 2025 – 2026  
Directed Taegon Lee's research on field theory of quantum Hall hierarchies, resulting in a first-author paper accepted as a Letter in [Physical Review B](#) on which I served as corresponding author
- Teaching Assistant, Department of Physics, POSTECH** 2021 – 2022  
Introduction to Relativity; Physics Laboratory I; Quantum Mechanics I
- Undergraduate Peer Mentor, POSTECH** 2019 – 2020  
General Physics I, General Physics II
- Residential Advisor, POSTECH** 2019 – 2020

## Academic Service

---

- Referee** *SciPost Physics* and *Physical Review B*
- Organizer** Condensed Matter Physics Journal Club at POSTECH